



LUNDS
UNIVERSITET

Matematikcentrum

Matematik NF

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MATP32

Assignment

1. (Centered maximal functions over balls and cubes)(4p)

Let v_d denote the volume of the unit ball in $\mathbb{R}^d$.

For $f \in L^1_{\text{loc}}(\mathbb{R}^d)$, define its maximal functions $M_c(f)$ and $M'_c(f)$ by

$$M_c(f)(x) := \sup_{r>0} \frac{1}{v_d r^d} \int_{|x-y|<r} |f(y)| dy \quad (x \in \mathbb{R}^d)$$

and

$$M'_c(f)(x) := \sup_{r>0} \frac{1}{r^d} \int_{x+(-r/2, r/2)^d} |f(y)| dy \quad (x \in \mathbb{R}^d),$$

respectively.

a) Show that $M'_c(f) \leq d^{d/2} \cdot v_d \cdot 2^{-d} \cdot M_c(f)$.

b) Show that $M_c(f) \leq 2^d \cdot (v_d)^{-1} \cdot M'_c(f)$.

2. (A.e. convergence of averages and a stronger version of the Lebesgue Differentiation Theorem)(6p)

Let f be a real-valued function in $L^1_{\text{loc}}(\mathbb{R}^d)$.

a) Let q be a rational number. Prove that there exists an $A_q \subseteq \mathbb{R}^d$ such that $|A_q| = 0$ and

$$\lim_{r \rightarrow 0^+} \frac{1}{|B(x, r)|} \int_{B(x, r)} |f(y) - q| dy = |f(x) - q| \quad \text{for all } x \in \mathbb{R}^d \setminus A_q.$$

b) Prove that there exists an $A \subseteq \mathbb{R}^d$ such that $|A| = 0$ and

$$\lim_{r \rightarrow 0^+} \frac{1}{|B(x, r)|} \int_{B(x, r)} |f(y) - f(x)| dy = 0 \quad \text{for all } x \in \mathbb{R}^d \setminus A.$$

c) For $x \in \mathbb{R}^d$, let $Q_n(x)$ be the unique dyadic cube of sidelength 2^{-n} containing x . Prove that there exists an $A \subseteq \mathbb{R}^d$ such that $|A| = 0$ and

$$\lim_{n \rightarrow \infty} \frac{1}{|Q_n(x)|} \int_{Q_n(x)} f(y) dy = f(x) \quad \text{for all } x \in \mathbb{R}^d \setminus A.$$

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3. (The strong maximal function) (6p)

Let $\mathcal{R}$ denote the class of all open rectangles in $\mathbb{R}^2$ with sides parallel to the coordinate axes, namely $R \in \mathcal{R}$ if, and only if, there exist open intervals $I, J \subseteq \mathbb{R}$ such that $R = I \times J$.

For $f \in L^1_{\text{loc}}(\mathbb{R}^2)$, consider its strong maximal function $M_S(f)$ by

$$M_S(f)(x_1, x_2) := \sup_{\substack{R \in \mathcal{R}: \\ (x_1, x_2) \in R}} \frac{1}{|R|} \int_R |f(y_1, y_2)| dy_1 dy_2, \quad (x_1, x_2) \in \mathbb{R}^2.$$

a) Prove that M_S is not of weak-type $(1, 1)$.

Hint. For $\delta \in (0, 1)$, consider the function $f_\delta(x_1, x_2) := \delta^{-2} \chi_{(-\delta, \delta)}(x_1) \chi_{(-\delta, \delta)}(x_2)$ and show that $M_S(f_\delta)(x_1, x_2) \geq |x_1|^{-1} |x_2|^{-1}$ for $|x_1|, |x_2| > 2\delta$.

b) Prove that M_S is bounded on $L^p(\mathbb{R}^2)$ for all $1 < p \leq \infty$.

Hint. Use the boundedness of the maximal function M on $L^p(\mathbb{R})$ in each variable.

Please hand in via Canvas by Wednesday, October 7.

References

- [1] Duoandikoetxea, Javier. *Fourier analysis*. Translated and revised from the 1995 Spanish original by David Cruz-Uribe. Graduate Studies in Mathematics, 29. American Mathematical Society, Providence, RI, 2001. xviii+222 pp.
- [2] Fefferman, Robert. *Multiparameter Fourier analysis*. Beijing lectures in harmonic analysis (Beijing, 1984), 47–130, Ann. of Math. Stud., 112, Princeton Univ. Press, Princeton, NJ, 1986.
- [3] Grafakos, Loukas. *Classical Fourier analysis*. Third edition. Graduate Texts in Mathematics, 249. Springer, New York, 2014. xviii+638 pp.
- [4] Grafakos, Loukas. *Modern Fourier analysis*. Third edition. Graduate Texts in Mathematics, 250. Springer, New York, 2014. xvi+624 pp.